Lesson 13 Teacher's Key: A Hit That Reacts

001 Game Creator (Windows only) - Instructor Reference Only

VIDEO GAME CREATION

What This Lesson Teaches

Using the confirmed built-in Actor – Flash action and Spawn Actor for a self-managed hit reaction - this engine's version of this week's idea, with an honest, documented limitation around exact flash-then-destroy timing that the other seven engines don't share.

Level 1 - Full Worked Solution

- Actor – Flash is a self-contained action - it manages its own color, repeat count, and duration internally, unlike every other engine this week, which needed a manually-authored delay.
- Spawn Actor(Explosion) fires immediately and independently, exactly like every other engine's explosion this week - this part of the lesson is fully consistent across all 8 engines.
- The enemy's Destroy action is left in its existing position (unchanged from Lesson 7 onward) - this is a deliberate, documented simplification for this engine specifically, not an oversight.

Highlighted lines are new this week; everything else is code you already have.

```
enemy Actor Trigger: Collided with actor (revised) ->
  If(game_active = 1)
    Variable – Modify Value(score, Add, 10)
    Variable – Modify Value(enemies_remaining, Subtract, 1)
    Play Sound(sfx_hit.wav)
    Actor – Flash(White, Duration: 0.2)
    Spawn Actor(Explosion) at (this actor's position)
    If(enemies_remaining <= 0)
      If(wave >= total_waves)
        Variable – Modify Value(game_active, Set, 0)
        Play Sound(sfx_win.wav)
        Message Box("You Win! - Press R to Restart")
      Else
        Variable – Modify Value(wave, Add, 1)
        Variable – Modify Value(enemies_remaining, Set, 5)
        Play Sound(sfx_wave.wav)
        Call Custom Event(SpawnWave)
    Actor – Destroy/Revive(Destroy)
```



Everything in the top box happens the instant the hit is detected. Only the enemy's own removal - the very last step - waits for the pause to finish.

Why This Structure

- Unlike Godot/Unity's await/Coroutine or GameMaker/TIC-80's alarm/countdown split, 001 Game Creator's confirmed API doesn't include a verified general-purpose 'wait N seconds mid-trigger' action - rather than fabricate one, this lesson keeps Destroy unchanged and treats the explosion effect as the primary, always-reliable visual for this engine.
- This mirrors the same honest-uncertainty approach this course has used before (e.g. Lesson 10's choice of a verified manual restart over an unverified built-in one) - a real, if imperfect, engine limitation, clearly labeled as such rather than papered over.

Level 2 / Level 3 Talking Points

- Ask students on this engine to actually test whether the flash is visible in their installed version - some versions/settings may show a brief flash even with an immediate Destroy, others may not; either result is expected and fine for Level 1.
- For advanced students: if their version DOES expose a Timer or delay mechanism (check the Actions list together), this is a great Level 3 stretch to hold Destroy back briefly, matching what every other engine's Level 1 already does.

Level 2 - complete code:

```
// Level 2 is verification only - the Level 1 code already produces a
working flash and explosion, so no lines change.
// Checks: play a full round and confirm every single hit flashes and
```

```
explodes, with no enemy left stuck mid-flash and no explosion left on
screen;
// confirm score/enemies_remaining/wave-clear/win-check all still happen
at exactly the same instant they did before this lesson - only the enemy's
own final removal is delayed;
// confirm a second bullet hitting an already-flashing enemy does not
double-register a kill.
```

Level 3 - complete code (all three goals combined):

Highlighted lines are the Level 3 additions on top of the Level 1 code.

```
enemy Actor Trigger: Collided with actor - Level 3 (holds Destroy back
briefly so Flash is fully visible):
  If(game_active = 1)
    Variable – Modify Value(score, Add, 10)
    Variable – Modify Value(enemies_remaining, Subtract, 1)
    Play Sound(sfx_hit.wav)
    Actor – Flash(White, Duration: 0.2)
    Spawn Actor(Explosion) at (this actor's position)
    If(enemies_remaining <= 0)
      (win/wave-advance logic, unchanged from Level 1)
    Timer – Start(0.2 seconds) -> On Complete: Call Custom
Event(FinishEnemyDestroy)

Custom Event: FinishEnemyDestroy ->
  Actor – Destroy/Revive(Destroy)
(Actor – Destroy/Revive(Destroy) moves out of the collision Trigger and
into this new Custom Event, so it fires after the Timer instead of in the
same instant as Flash.)

Explosion Actor Template's own Trigger: On Create -> (Level 3 two-frame
animation)
  Timer – Start(0.15 seconds) -> On Complete: Actor – Change
Graphics(Explosion_Bright)
```

Common Student Mistakes

- Assuming the flash isn't working at all when it's actually just being cut short by the immediate Destroy - the explosion effect, not the enemy's own flash, is what to check first when troubleshooting this engine.
- Forgetting to assign a Graphics asset to the new Explosion Actor Template, resulting in an invisible (but otherwise functioning) spawned Actor.
- Not setting a Lifetime/Duration on the spawned Explosion (if available), leaving them on screen indefinitely across a long playthrough.